Submission Summary

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Paper Title

E-MARS: Emergent Modular Adaptive Reliability System for Microservice Architectures

Abstract

Contemporary distributed systems are progressively utilizing microservice architectures that consist of many loosely connected services. Although this approach enhances scalability and speeds up development, it brings notable difficulties in managing reliability, diagnosing failures, and optimizing resources. Conventional Site Reliability Engineering (SRE) methods function at the single service level with fixed monitoring and rule-driven policies, frequently neglecting the intricate interdependencies present in extensive systems.

To overcome these constraints, we introduce E-MARS (Emergent Modular Adaptive Reliability System), a data-centric framework that facilitates module-sensitive reliability management. E-MARS represents microservice ecosystems as weighted interaction graphs based on observability data, encompassing metrics, traces, and service communication relationships. Utilizing techniques for maximizing modularity, the system autonomously discovers communities of closely linked services, known as emergent modules, and considers them the primary units for managing reliability.

We introduce a mathematical formulation of the framework, encompassing the modularity objective function, and create a scalable pipeline for detecting communities in extensive microservice environments. Testing on synthetic and simulated workloads shows that module-aware control notably decreases end-to-end latency, shortens mean time to recovery (MTTR), reduces error propagation, and enhances resource utilization relative to traditional per-service approaches.

These results suggest that emergent modularization—previously noted in deep neural networks—can function as a useful organizing strategy for adaptive reliability management in contemporary cloud-native systems

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Submission Files

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Submission Questions Response

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